convalescence as a white dwarf star. And this pressure comes directly from quantum mechanics.

A QUANTUM HEAP

The first thing you need to know is that most of the subatomic particles you know and love—electrons, protons, neutrons, neutrinos, quarks—are fermions, which, in this context, means they are fiercely independent, in a particle physics kind of way. Specifically, they obey the *Pauli exclusion principle*, which says that they won't abide being in the same place and the same energy state at the same time. This is why, if you recall high school chemistry lessons, electrons attached to atoms end up in different kinds of "orbitals," which are really just energy states.

Anyway, in the core of a burned-out, collapsing star, there are so many atoms, pressed so tightly together, their electrons start to get antsy. At those kinds of pressures, the electrons aren't bound to specific atoms, but rather are packed in together in a big atomic mess so crowded that they have to jump to higher and higher energy states to keep from all being in the same one. This creates a kind of pressure, called *electron degeneracy pressure*, which is strong enough to halt the collapse of the star and create an entirely new kind of object: a white dwarf.

A white dwarf is a kind of star that isn't burning at all. It has no fusion. It is a solid object held up entirely by the quantum mechanical principle that electrons just don't like each other that much. And it can persist, silently smoldering, for billions and billions of years, until it slowly fades and cools and darkens, and is disintegrated in the Heat Death of the universe, ignited in the Big Crunch, or torn apart by phantom dark energy in the Big Rip, along with everything else.

Unless it gets just a little bit more mass.

Electron degeneracy pressure can do a lot. It can support an ENTIRE STAR. But only up to a point. If something happens to push the white dwarf past this point—it pulls in material from a companion star, or collides with another white dwarf—it will have too much mass for the degeneracy pressure to balance further collapse. Once that balance is tipped, a number of things happen in rapid succession.

The central core temperature of the star increases. Carbon begins burning. The material of the star starts to roil and churn, dragging more